

Traffic pollution contributes to more than 1,800 premature deaths per year, study estimates

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ABC Science

Air Pollution

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Traffic pollution is linked to heart and lung disease, chest infections, asthma, diabetes, and many other health issues. (Getty: Peter Macdiarmid)

In short:

The first assessment of the health impact of traffic-related air pollution in Australia estimates it contributes to more than 1,800 premature deaths each year.

This is significantly less than a 2023 estimate of more than 11,000 deaths, but still higher than the annual road toll.

What's next?

The study's authors say the findings show the need for a rapid transition to electric vehicles, which emit less pollution, as well as better public transport infrastructure.

Vehicle pollution may contribute to more deaths each year than the number caused by road accidents.

That's according to the first assessment of the health impact of traffic-related pollution (TRAP) in Australia.

The new research, led by the University of Tasmania's Centre for Safe Air, estimates particulate and nitrogen dioxide pollution from traffic leads to more than 1,800 earlier than expected deaths a year, or more than the 2024 road toll of 1,300 fatalities.

The study found cars were responsible for about half of the estimated total annual premature deaths from air pollution, which is Australia's leading environmental cause of premature death and disease.

The research, [published in the journal Environmental Research](#), estimates premature deaths from TRAP are probably lower than a 2023 University of Melbourne estimate of 11,000 per year.

Fay Johnston, director of the Centre for Clean Air and a co-author of the study, said the new research showed "traffic pollution is the single-most important source of air pollution."

"If we're wanting to reduce chronic heart disease and illness and deaths, traffic pollution is a good place to look."

Professor Johnston said the new study's main finding was conservative, partly to avoid uncertainty.

If the underlying assumptions of how pollution exposure impacts health were tweaked to be comparable with studies conducted in New Zealand, the figure would be more than 8,000 annual deaths.

"It's an example of how difficult it is to attribute mortality at a population level," Professor Johnston said.

Quantifying the health impact of traffic pollution

Vehicles mostly contribute to air pollution through the emission of particulate matter (PM) and nitrogen dioxide (NO₂) in tailpipe exhaust, or by tyre and brake wear.

These pollutants cause inflammation, contribute to heart and lung disease, and are linked to other health problems such as diabetes and asthma.

Others studies in Australia have estimated the health impact attributable to PM, but not NO₂.

The authors of the latest study used a combination of air quality data from 2015 and statistical modelling to estimate the concentrations of TRAP across Australia.

They then matched this pollution-exposure map with population data and health impact assessments from epidemiological studies to estimate the overall number of premature deaths.



New South Wales, Victoria and Queensland accounted for more than three quarters of the premature deaths attributable to air pollution in general.

The two most populous states, New South Wales and Victoria, accounted for most of the premature deaths specifically attributable to traffic-related air pollution.



"That's reflecting Sydney and Melbourne," Professor Johnston said.

"It's simply due to numbers of people. There's millions of people living in those great big cities."

Why the premature deaths figure is lower than previous estimates

In 2023, University of Melbourne research hub Melbourne Climate Futures (MCF) [estimated TRAP contributed](#) to more than 11,000 annual deaths in Australia.

However, unlike with the latest study, this figure was not arrived at through peer-reviewed research.



A mix of cars and trucks fill four lanes of peak hour traffic on freeway.

Traffic congestion in Australian cities is generally getting worse, due to more cars using the roads.
(ABC News: Andrew O'Connor)

The MCF researchers took the results of a [2016 New Zealand health study](#) and extrapolated them to Australia's larger population, as part of a [position statement](#) endorsed by major non-profit health organisations such as the Cancer Council and the Heart Foundation.

Clare Walter, a health and policy researcher at the University of Melbourne who co-authored MCF's 2023 position statement, told the ABC the estimate of premature deaths in the recent Centre for Safe Air study was too conservative.

"And that worries me because it will feed into cost-benefit analyses that guide policy decisions," she said.

Unlike in the NZ study (which informed the MCF estimate), the Centre for Safe Air researchers applied a minimum threshold for exposure to air pollution, which they said was to account for naturally occurring levels of air pollution and because of uncertainty around the health impact.

Incremental increases in pollution exposure at low levels result in a greater difference in health impact than the same-size increases at higher levels.

As a result, determining where to place a minimum threshold for health impact, or whether to have any threshold at all, significantly affects the overall health impact calculation.

World Health Organization air quality guidelines state there is no safe level of exposure to air pollution.

Dr Walter said "the evidence very clearly shows that there is no threshold for a mortality effect".

"I consider the [NZ study] risk estimates to be more relevant for Australia."

 A Transperth train travels from left to right with blurry cars heading in the opposite direction in the foreground.

A single passenger train can replace hundreds of cars, resulting in less traffic pollution overall. (*ABC News: Eliza Laschon*)

Simon Hales, an environmental epidemiologist at the University of Otago and co-author of the NZ research from 2016, said the use of a minimum-exposure threshold in the new study affected its premature deaths estimate.

"We didn't assume any [minimum] threshold. We assumed air pollution had a health effect right down to zero exposure," Dr Hales said.

"If they had done that, they would have found air pollution had a much greater effect."

Professor Johnston said the Centre for Safe Air "used the best available figures from studies all around the world" for a "rigorous, defensible study."

"The key thing is we wanted to be transparent and show it all, that's why the paper shows estimates from a whole range of modelling," she said.

Deeper dive needed for certainty

Dr Hales said Australia needed more intensive research into the health impact of air pollution.

The Centre for Safe Air study relied upon epidemiological studies from other countries, including NZ, to estimate the health impact of exposure to air pollution.

Australia does not have such a study, which would require trawling through health records to attribute mortality and morbidity to exposure to air pollution.

"What's missing in Australia is local evidence of the sensitivity of the population [to air pollution]," Dr Hales said.

"These studies are not easy to do, because we know the long-term effects of exposure to air pollution are very much greater than the short-term effects."

Dr Walter said Australia need a long-term epidemiological study focusing on air pollution.

"Instead of debating whose estimate is 'more accurate', I think the best path forward is to follow New Zealand's example and carry out a thorough long-term study here in Australia that produces risk estimates informed by the Australian population."

Calls to reduce traffic pollution

Professor Johnston said the study showed the need for a rapid transition to electric vehicles, which emit less pollution (they still emit some pollution through tyre and brake pads).

"We also need to support public transport and active transport [such as walking and cycling] to

reduce emissions and protect health."

Doctors for the Environment Australia air pollution spokesperson Vicki Kotsirilos, who is a GP and medical practitioner, said a public information campaign about the harms caused by traffic pollution would help reduce the health impact.

"Just like decade ago we had a very strong campaign about the harms of smoking," Dr Kotsirilos said.

"We live in a country where we are very reliant on our vehicles.

"If parents knew idling a car at childcare drop-off centres increases the risk of childhood asthma they would be sure to turn their car off."

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